

AWS VPC Design & Operational Networking Standards

Nexlify Technologies | Region ap-south-1 | Primary AZ ap-south-1a | Secondary AZ ap-south-1b

1. Default VPC Lockdown Policy

No compute, database, or network interface resource is permitted inside the AWS default VPC of any Nexlify account. This policy is enforced by an AWS Organizations Service Control Policy (SCP) named 'DenyDefaultVPCResources'. The SCP denies CreateNetworkInterface, RunInstances, CreateDBInstance, CreateFunction, and related actions when the target subnet belongs to a VPC whose isDefault attribute is true. Existing default VPCs remain for AWS service compatibility but are treated as quarantined. Platform Engineering runs a weekly Config rule evaluation to detect any residual ENIs or security groups attached to default VPC subnets and opens auto-tickets for cleanup. Teams that accidentally target the default VPC receive an automated Slack notification in #platform-alerts with remediation steps pointing to the approved operational VPC Terraform module.

2. Current Operational VPC Inventory

Production workloads reside in VPC vpc-0a1b2c3d4e5f67890 (CIDR 10.40.0.0/16) named 'nexlify-prod-ap-south-1'. Non-production (dev/staging) workloads use VPC vpc-0f9e8d7c6b5a43210 (CIDR 10.50.0.0/16) named 'nexlify-nonprod-ap-south-1'. Shared services (Transit Gateway attachments, VPC endpoints, Route 53 Resolver) live in VPC vpc-0123456789abcdef0 (CIDR 10.30.0.0/16) named 'nexlify-shared-ap-south-1'. All three VPCs are created from the same Terraform module version 4.7.2. Internet Gateway and NAT Gateway placement prefers ap-south-1a to reduce cross-AZ data transfer costs for outbound traffic. One NAT Gateway per AZ is provisioned for production; non-prod uses a single NAT Gateway in ap-south-1a only.

3. Subnet CIDR Bifurcation Design (Non-Prod Example)

For the non-production VPC (10.50.0.0/16) the following four subnets are mandated. Public subnets receive a total of 10 /24 blocks conceptually split as 5+5 usable host capacity planning, while private subnets receive the majority of address space. AZ preference: 50% of capacity in ap-south-1a (primary) and 40% in ap-south-1b (secondary), with the remaining reserved for future AZ expansion. Concrete allocations used by Platform Engineering:

Subnet Name	CIDR	AZ	Type	Purpose / Notes
subnet-nonprod-pub-1a	10.50.0.0/22	ap-south-1a	Public	ALB, NAT GW, bastion (primary)
subnet-nonprod-pub-1b	10.50.4.0/22	ap-south-1b	Public	ALB secondary, failover
subnet-nonprod-pvt-1a	10.50.16.0/20	ap-south-1a	Private	EKS nodes, Redis, RDS primary
subnet-nonprod-pvt-1b	10.50.32.0/20	ap-south-1b	Private	EKS nodes secondary, replicas

Public subnets are sized at /22 (1022 usable IPs each) to accommodate multiple load balancers and future ingress controllers while keeping the public footprint small. Private subnets use /20 (4094 usable IPs) because EKS node groups, Karpenter provisioned nodes, and stateful services consume far more addresses. The 50/40 split (ap-south-1a vs ap-south-1b) is intentional: most stateful pods and PVCs are scheduled into ap-south-1a via topology constraints so that inter-node traffic stays inside the same AZ, reducing data transfer charges that would otherwise accumulate at \$0.01/GB cross-AZ. Route tables for private subnets point 0.0.0.0/0 to the NAT Gateway in the same AZ when available; otherwise they fall back to the primary NAT in ap-south-1a.

4. Cost-Optimised AZ Preference Rule

Whenever a resource supports AZ selection, prefer ap-south-1a. Examples: EKS control plane endpoint placement, RDS primary instance, ElastiCache primary node, MSK broker 1, and Karpenter NodePool topology spread constraints. Cross-AZ traffic is unavoidable for Multi-AZ databases, but application-to-database latency and cost are minimised by co-locating the majority of pods with the primary. The Platform team monitors the AWS Cost Explorer data transfer line item weekly; any unexplained spike triggers a review of recent subnet or nodepool changes.

5. VPC Endpoints & Private Connectivity

Interface endpoints for ECR, S3 (gateway), STS, EC2, EKS, CloudWatch Logs, and Secrets Manager are deployed in both private subnets of every operational VPC. This eliminates NAT Gateway charges for AWS API traffic and keeps management plane traffic off the public internet. Endpoint policies restrict access to Nexlify account principals only. Transit Gateway attachments connect the three VPCs and the on-premises data centre via Direct Connect; route propagation is controlled by the Network team.

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